

1 Covariance Analysis in Health Insurance

Given,

$$x = 3.2, 4.0, 5.5, 4.8$$

$$y = 15, 18, 25, 20$$

Step 1: Mean

$$\text{Mean of } x = (3.2 + 4.0 + 5.5 + 4.8) / 4 = 4.375$$

$$\text{Mean of } y = (15 + 18 + 25 + 20) / 4 = 19.5$$

Step 2: Covariance

$$\text{Covariance} = 5.31 \text{ (approx.)}$$

Conclusion: Since the covariance is positive, the premium increase and claim frequency have a positive relationship. This means when the premium increase is higher, the number of claims also tends to increase.

2 Data Smoothing using Equal-Frequency Binning

The 27 blood pressure values are divided into 3 bins with 9 values each.

Bin 1:

$$110, 112, 115, 115, 118, 120, 120, 122, 124$$

Bin 2:

$$124, 126, 126, 126, 126, 130, 135, 135, 138$$

Bin 3:

$$138, 138, 138, 140, 145, 150, 152, 160, 180$$

Conclusion:

Equal-frequency binning groups the data into bins containing the same number of values, helping to smooth small fluctuations.

3 Equal-Width Binning for Risk Categorization

* Minimum BMI = 18.5

* Maximum BMI = 39.5

* Range = 21

* Bin width = $21 \div 3 = 7$

Bins:

* Bin 1: 18.5 - 25.5

* Bin 2: 25.5 - 32.5

* Bin 3: 32.5 - 39.5

Conclusion:

Equal-width binning divides the data into intervals of equal size for easy analysis.

4 Data Reduction using Bin Medians

Divide the sorted cost data into 3 equal-frequency bins.

Median of Bin 1 = ₹ 145

Median of Bin 2 = ₹ 155

Median of Bin 3 = ₹ 180

Conclusion:

Replace all values in each bin with the bin median.
This reduces data variation while preserving the overall trend.

5 Standardization of Patient Age Data

Ages: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Mean = 33

a) Mean Absolute Deviation (MAD)

MAD = 9.8

b) Z-scores (first four ages):

* Age 18 $\rightarrow$ -1.13

* Age 22 $\rightarrow$ -0.79

* Age 25 $\rightarrow$ -0.54

* Age 42 $\rightarrow$ 0.76

Conclusion:

Standardization converts the data into z-scores so that values can be compared on the same scale.